MIL-B-49430/12A(ER)
15 June 1987

SUPERSEDING
MIL-B-49430/12(ER)
21 August 1986

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5093/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

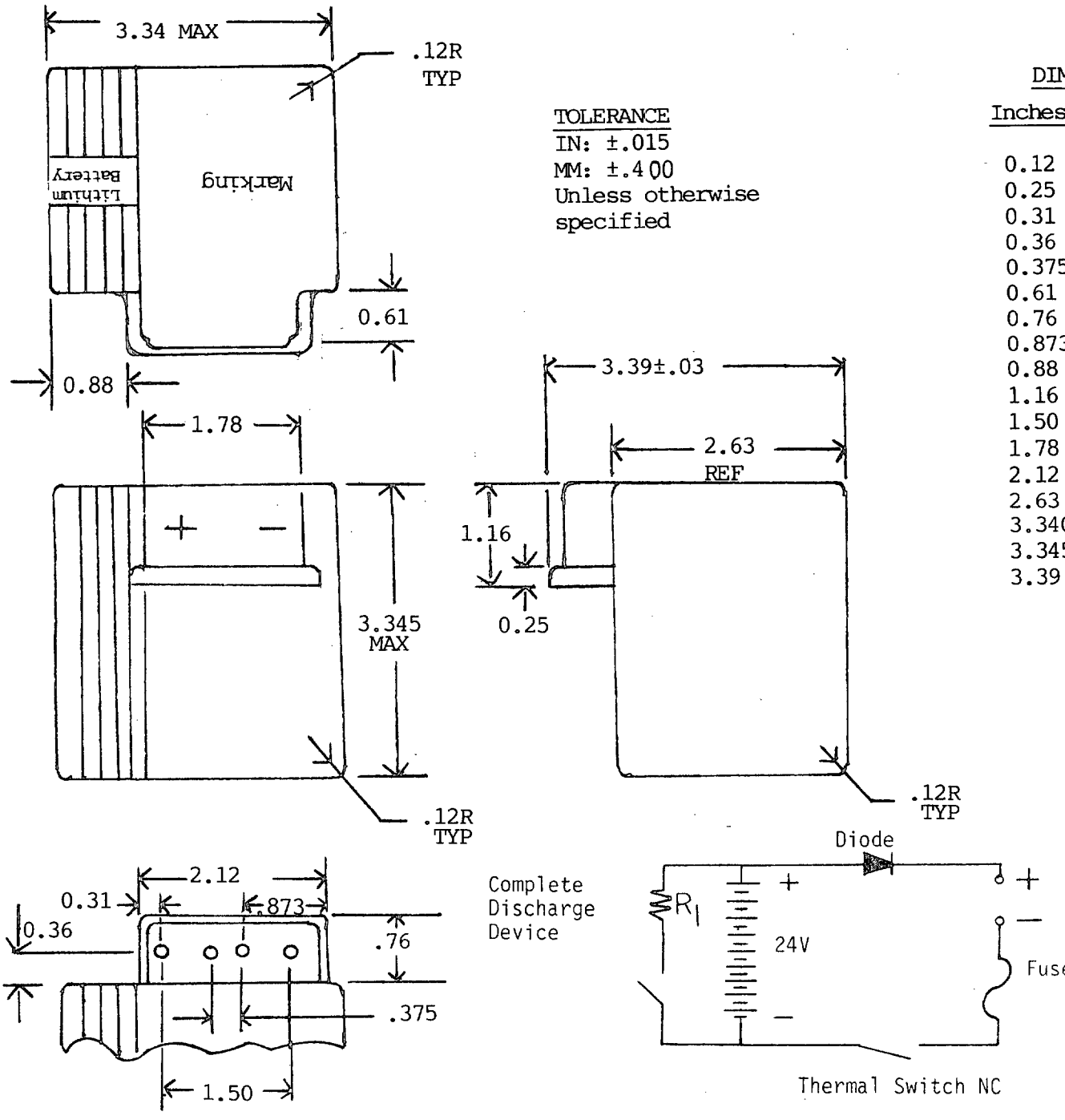
The requirements for acquiring the battery described herein shall consist of this specification and the latest issue of MIL-B-49430(ER).

FSC 6135

AMSC N/A

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TOLERANCE
IN: ±.015
MM: ±.400
Unless otherwise specified

DIMENSIONS

Inches	MM
0.12	3.05
0.25	6.35
0.31	7.87
0.36	9.14
0.375	9.53
0.61	15.49
0.76	19.30
0.873	22.17
0.88	22.35
1.16	29.46
1.50	38.10
1.78	45.21
2.12	53.85
2.63	66.80
3.340	84.84
3.345	84.96
3.39	86.11

FIGURE 1. Dimensions and configuration.

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NOTES:

1. All dimensions are in inches.

2. Socket wiring schematic to be printed on jacket in suitable location near the socket on the terminal face of the battery.

3. The battery connector shall be a flat, fiberglass socket connector with two (2) push-in terminals per MIL-B-18 as shown in Figure 1.

4. R_1 value shall be between 460 ohms and 650 ohms and have a power rating of 2.0 watts.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Maximum voltage: 27.45 volts.

Minimum voltage(for capacity): 18.00 volts.

The battery shall contain a 1.25 ampere non-replaceable slow blow fuse in the negative leg.

The battery shall contain a diode wired in series. See paragraph 3.5.5.2 of the Basic Specification.

A thermal switch shall be located within the nine cell cluster in accordance with paragraphs 3.5.5 and 3.5.5.1 of the basic specification

Terminals: Two (2) hole flat plate type (with lead-in countersink on both holes) female socket per MIL-B-18, Type IV, with no obstruction of any holes.

Weight (maximum): 1.43 pounds (650 grams).

Capacity Tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in Hours to 18.00 Volts
I	3.4
L	1.7
H	6.8
HT	5.5
LT	1.6
T	3.1
D	3.1
ILR	1000

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Initial Voltage Delay: During the I, H, HT, T, D, and ILR discharge tests, initial voltages below the 18.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests, initial voltage below 18.00 volts cannot exceed a 60.0 second duration.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in para 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS

Capacity Tests:

(1) Storage: Details on storage conditions for all capacity tests are described in the basic specification. (see 4.8.9).

(2) Discharge:

Cells: Discharge at 1.0 amp constant current at 70 ± 5°F for a minimum of 3.1 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 0.5 ampere constant current for 6.2 hours into voltage reversal.

Batteries: For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at 1.0 ampere constant current to zero volts. For the H and HT capacity tests, the batteries shall be discharged through 1.0 ampere constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test, the batteries shall be force discharged at 0.5 ampere constant current for 6.8 hours immediately after zero volts is obtained. There shall be no venting, fire or explosion when the batteries are force discharged. For the ILR test, a fixed resistance of 8000 ohms shall be used to a cut-off voltage of 18.00 volts or 1000 hours.

Closed circuit voltage: Load resistance of 48 ohms shall be used between the battery terminals. Voltage shall be 18.0 volts or greater within five seconds.

Custodian:
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(Project 6135-A201-12)

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